SUNSTREAM AVIATION

CLEAN FUEL CLEANER SKIES

Ethanol to Jet Fuel

Joshua Olusoji Omar Zaruk

AGENDA

01 **Process Overview**

Jet Fuel (SAF) production & importance

02 **Process Description**

The Ethanol-to-Jet Process Explained

03 **Process Economics**

Production Costs, Income Analysis & NPV

04 **Conclusion**

Final Recommendations and Q&A!

01 Process Overview

Why SAF and How?

The Global Warming Problem

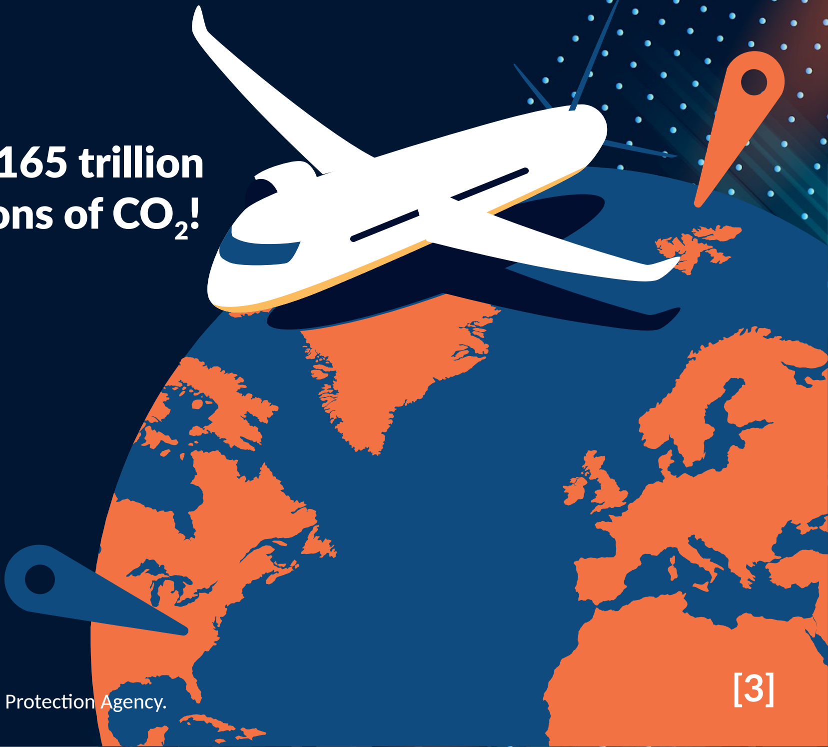
29%

Of all green house gas emissions are due to transportation

Approx. 165 trillion metric tons of CO_2!

9%

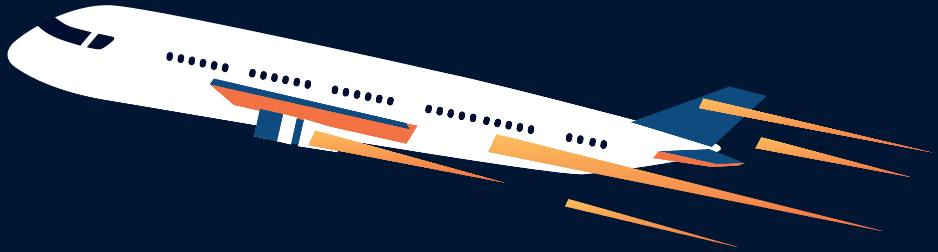
Of which is via aircraft!



[1] United States Environmental Protection Agency. *Sources of Greenhouse Gas Emissions*. United States Environmental Protection Agency. https://www.epa.gov/ghgemissions/sources-greenhouse-gas-emissions.

2050

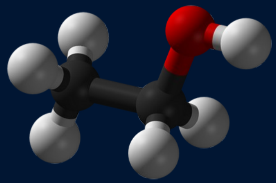
Grand Challenge: Completely replace all Jet A fuel with SAF domestically



Aviation is **exceptionally hard** to decarbonize!
It's mobile, requires high-energy-density fuel,
and requires long lead times for innovation!

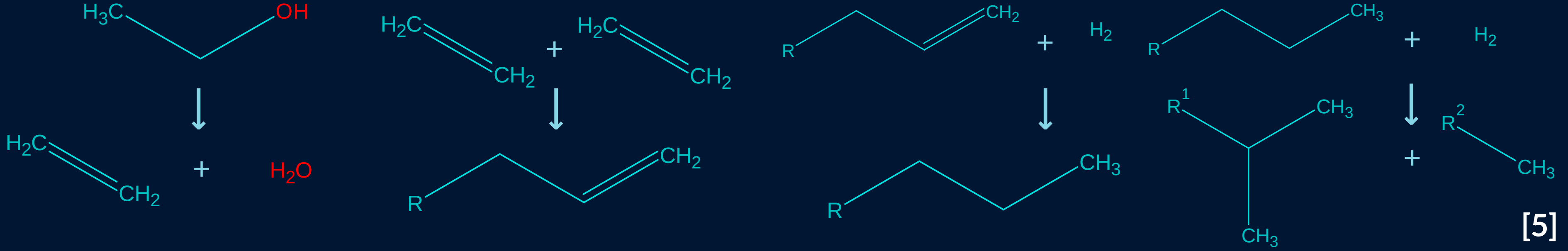
[4]

Chosen Process



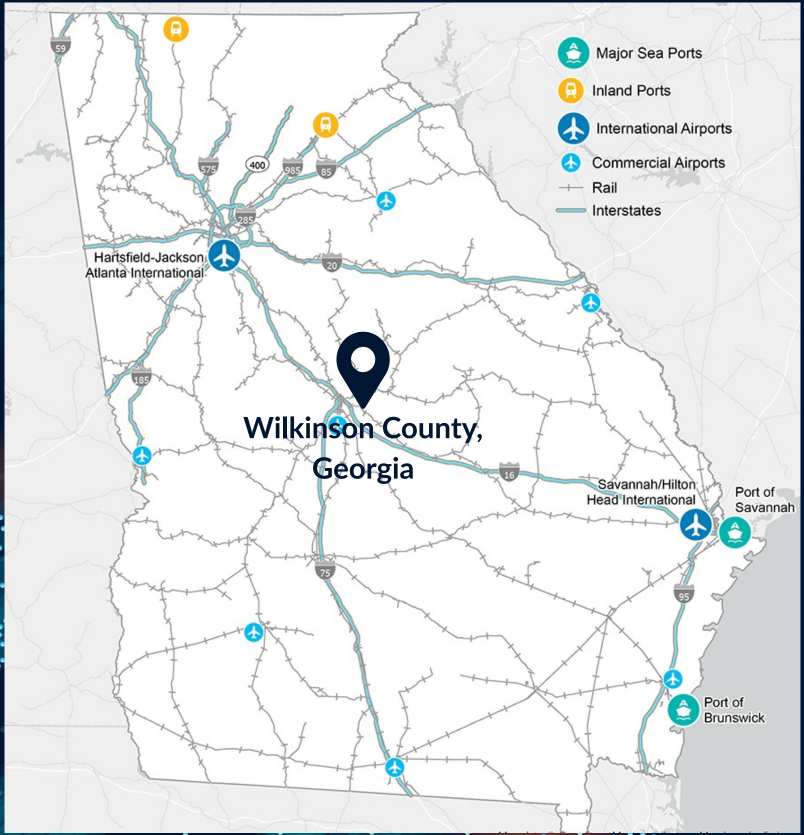
Ethanol → Jet Fuel

Dehydration → Oligomerization → Hydrogenation → Hydroprocessing



[5]

Plant Design Basis



Location: Wilkinson County, Georgia

In-state feedstock & access to Port Savannah for **international transactions**!

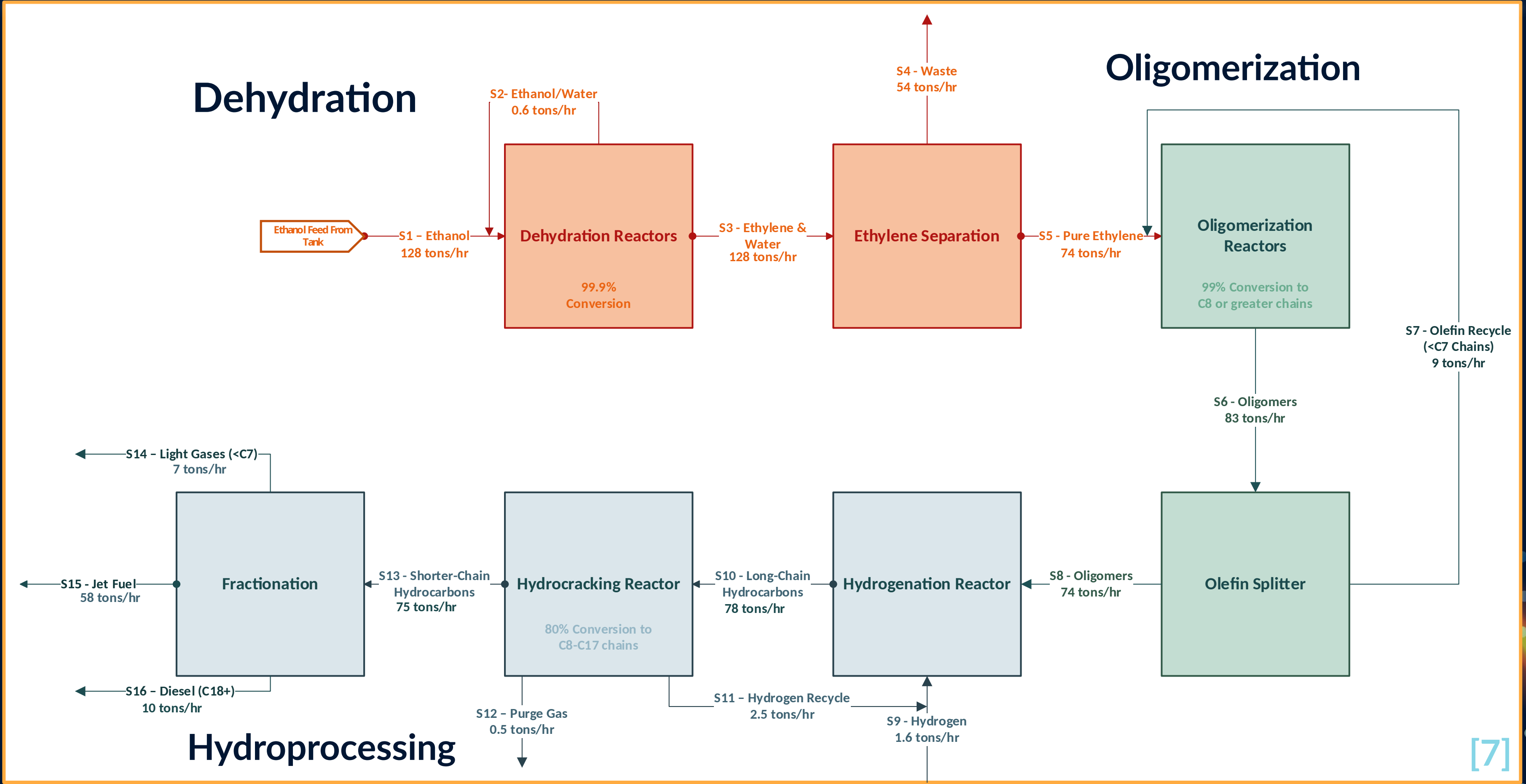
Green ethanol feedstock maximizes **RIN's & Tax Credits!**

Close proximity to Hartsfield-Jackson & Savannah Hilton airports!

Production: 10,000 Barrels/Day

This high daily production would **maximize profitability.**

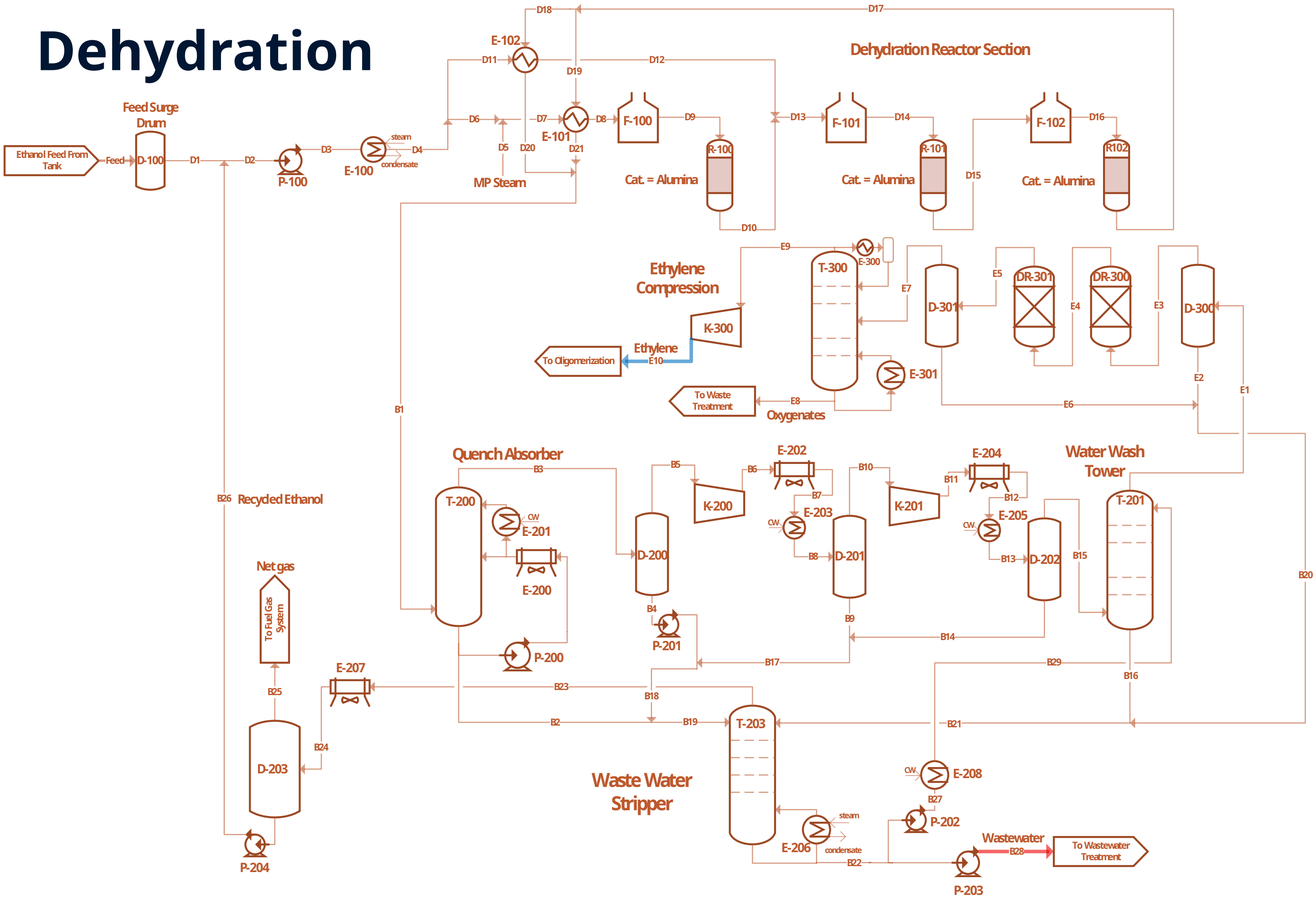
Block Flow Diagram & Material Balances



02 Process Description

The Ethanol-to-Jet Process Explained

Dehydration



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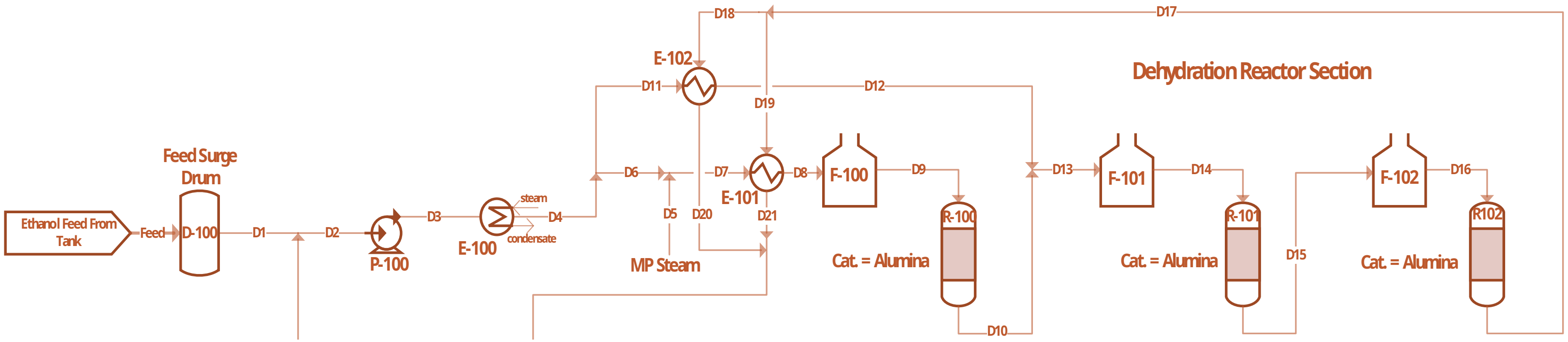
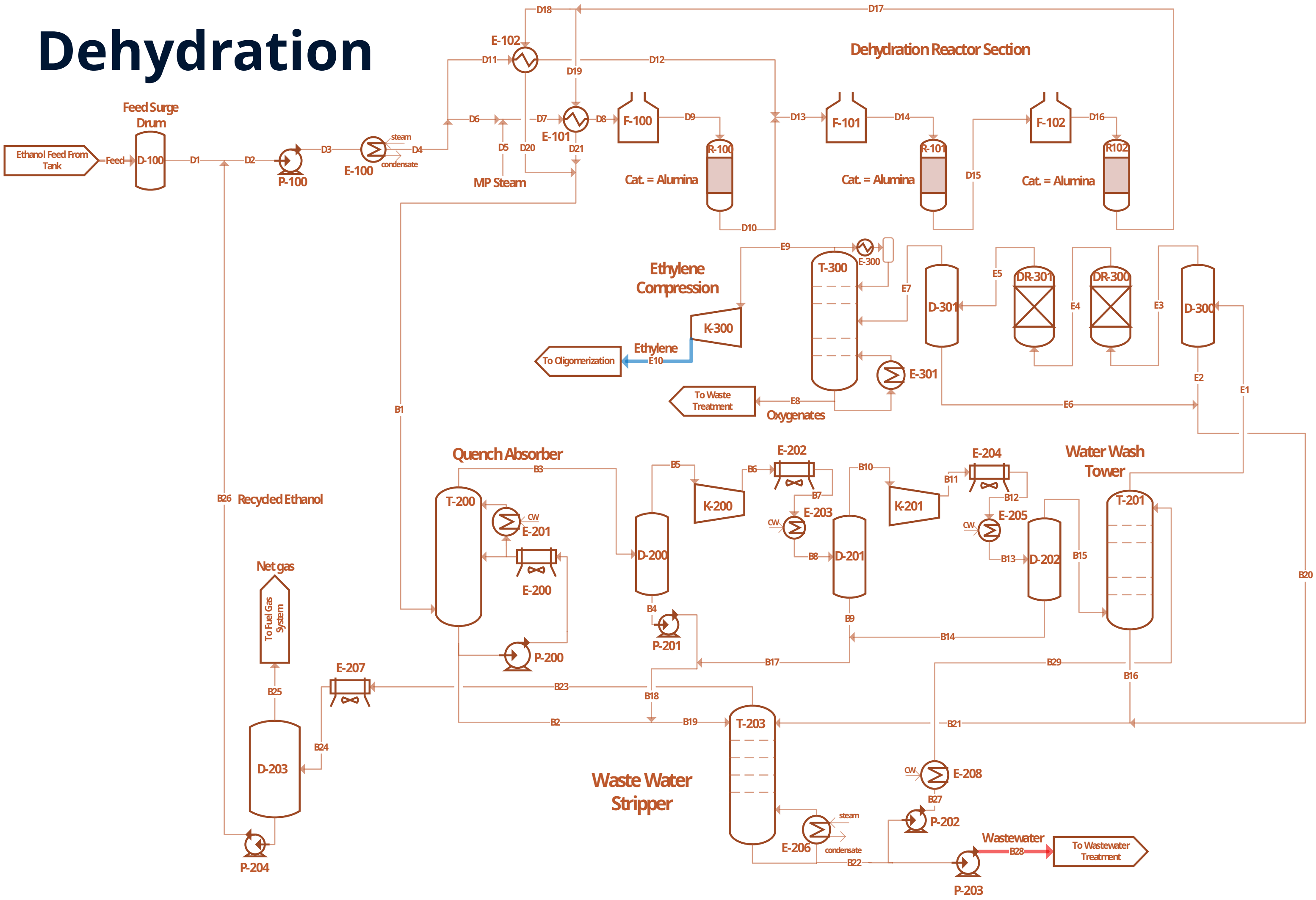
Dehydration Reaction



Component	Units	D11	D12	D13	D14	D15	D16	D17	D18	D19	D20	D21
Temperature	F	278	707	584	842	586	860	**837**	837	837	599	381
Pressure	psia	101	91	91	86	95	90	**85**	85	85	75	75
Total Mass Flow	lb/hr	128.727	128.727	441.242	441.242	441.242	441.242	**441.242**	220.621	220.621	220.621	220.621
Ethanol	lb/hr	122.526	122.526	123.751	123.751	1.238	1.238	**12**	6	6	6	6
Water	lb/hr	6.145	6.145	243.227	243.227	290.846	290.846	**291.323**	145.661	145.661	145.661	145.661
Ethylene	lb/hr	46	46	73.428	73.428	147.496	147.496	**148.227**	74.119	74.119	74.119	74.119

[10]

Dehydration



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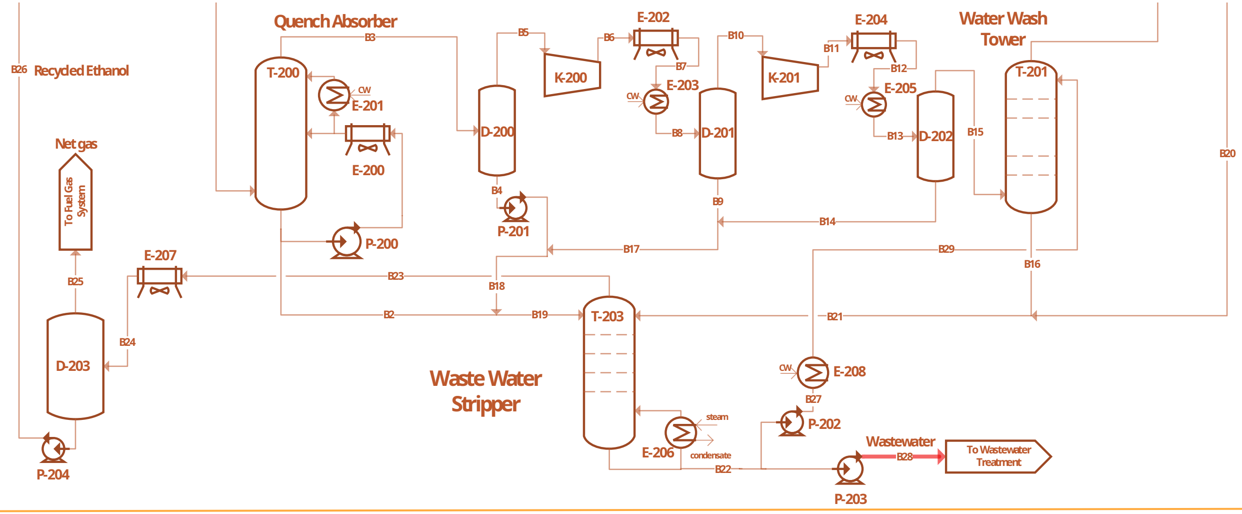
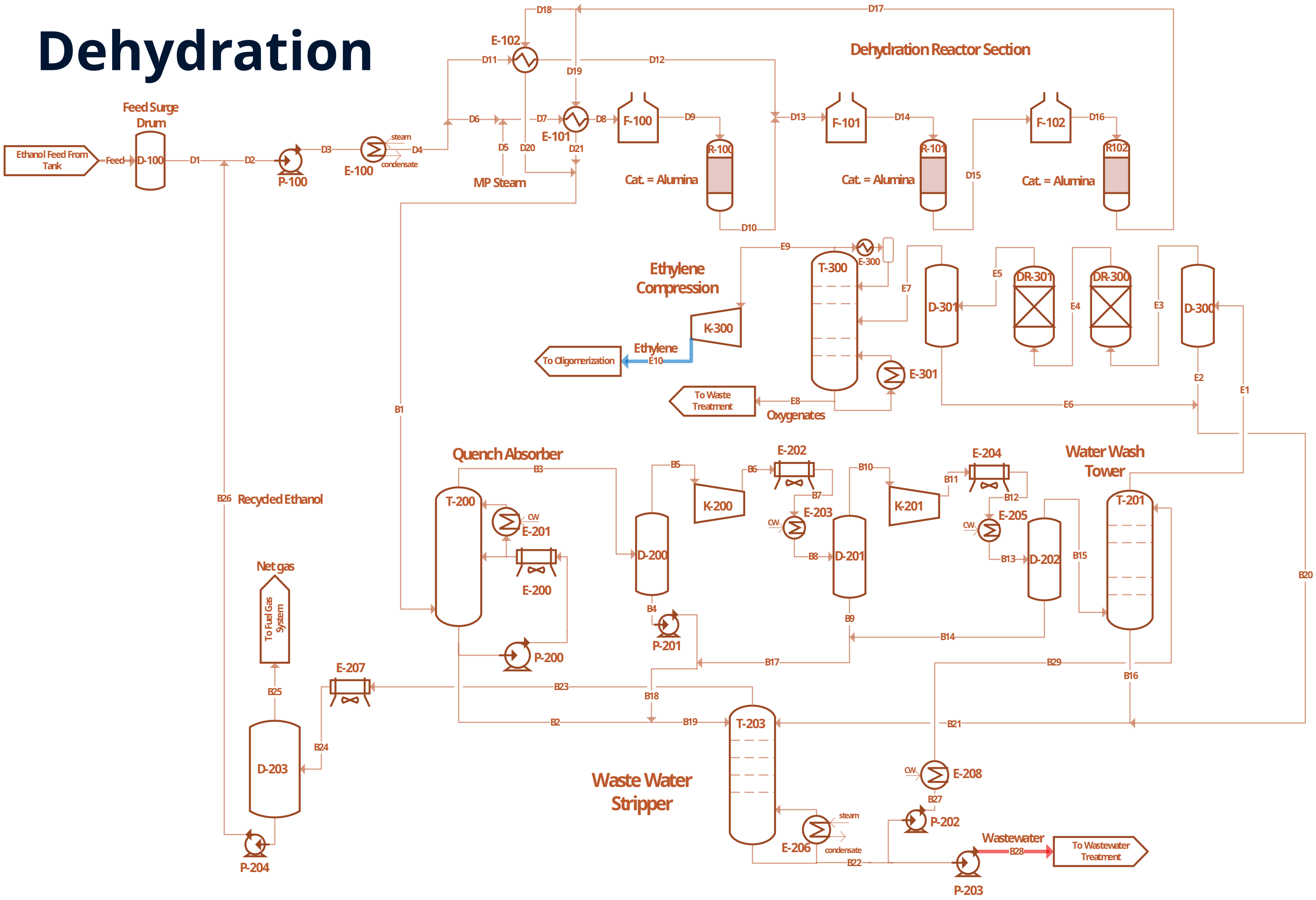
Byproduct Removal



Component	Units	B20	B21	B22	B23	B24	B25	B26	B27	B28	B29
Temperature	F	91	96	274	252	140	147	147	275	275	275
Pressure	psia	406	406	44	44	39	39	39	411	73	411
Mass Flows	lb/hr	1,523	12,607	477,948	1,765	1,765	636	1,129	11,023	466,924	11,023
Ethanol	lb/hr	6	7	2	6	6	2	5	2	2	0
Water	lb/hr										

[12]

Dehydration



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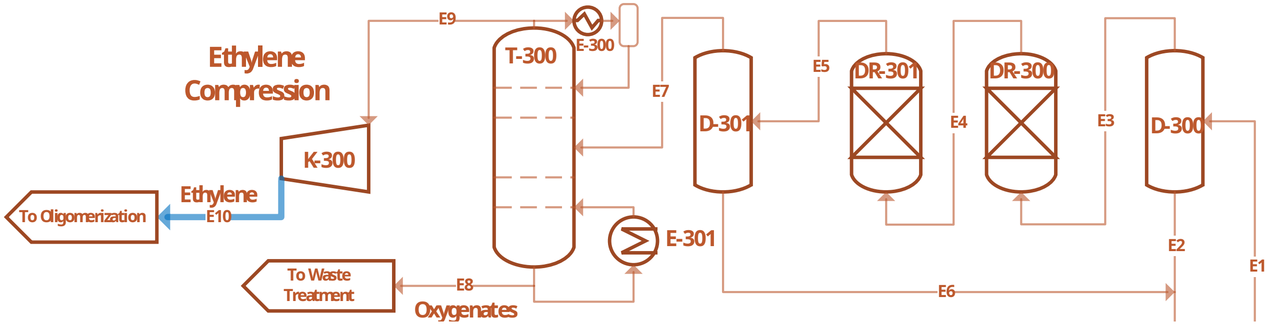
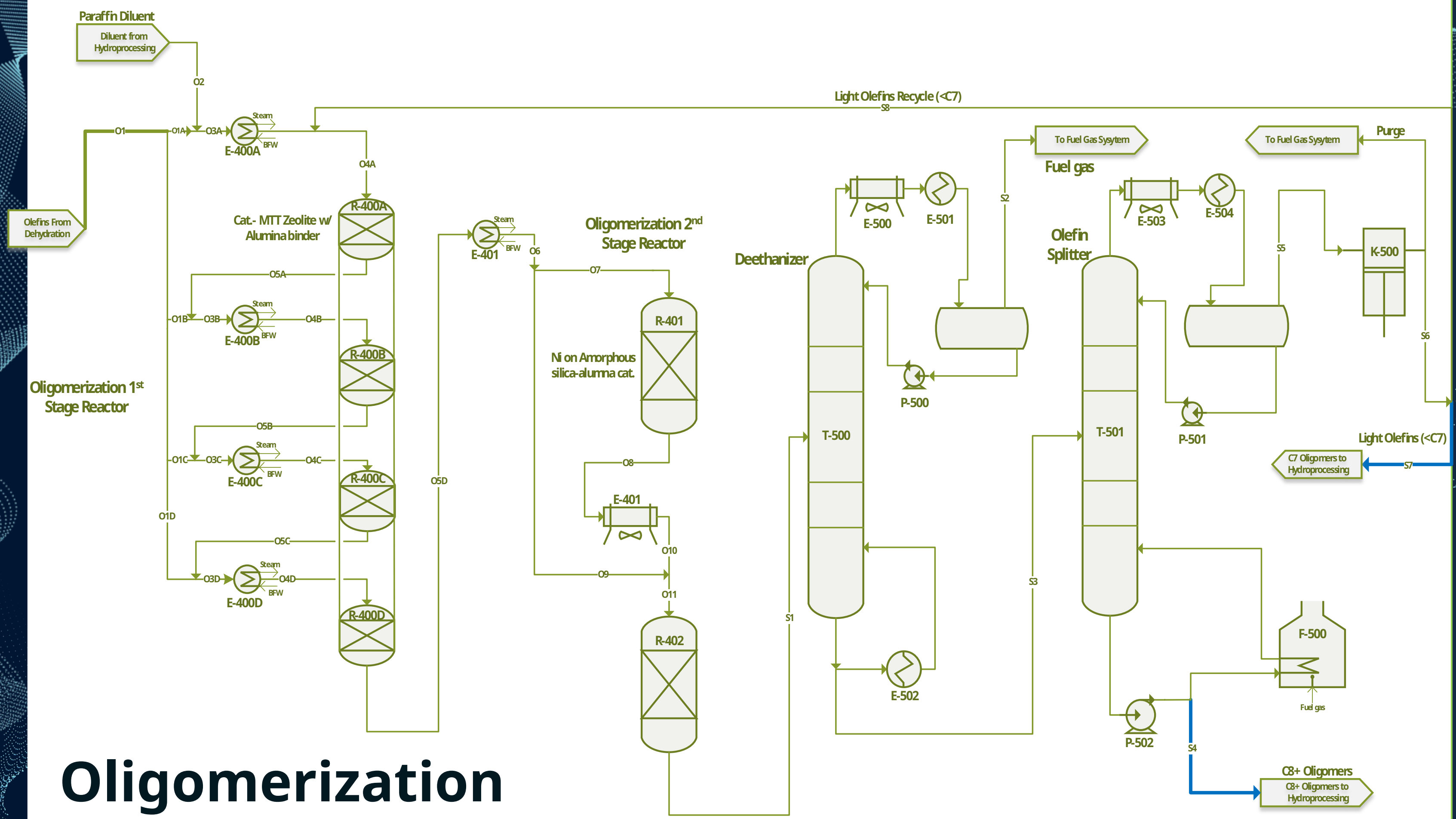
Ethylene Purification

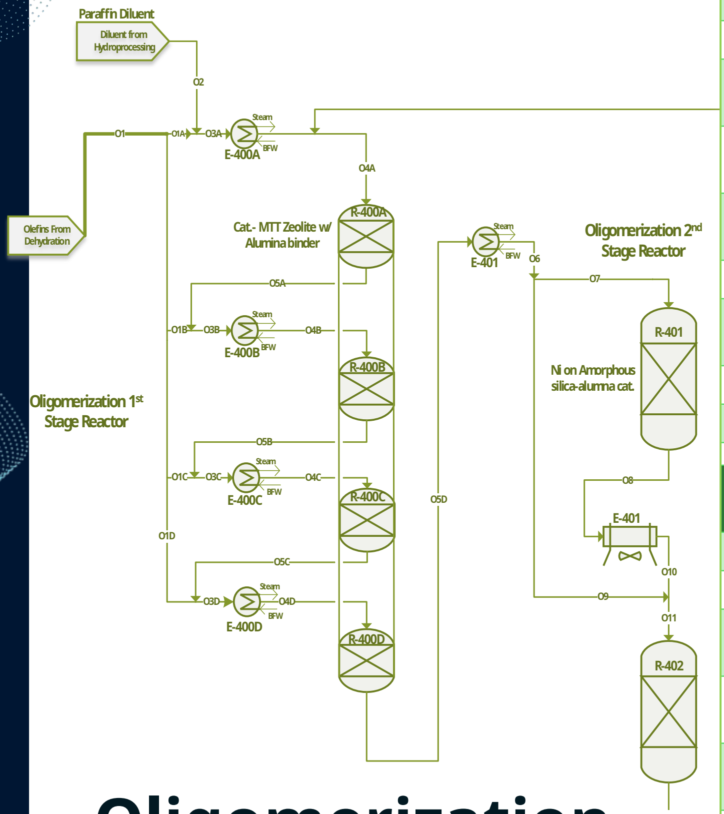


Component	Units	E1	E2	E3	E4	E5	E6	E7	E8	E9
Temperature	F	130	91	91	91	91	91	91	134	2
Pressure	psia	406	406	406	406	406	406	406	402	400
Mass Flows	lb/hr	150.522	1.425	149.096	148.999	148.999	98	148.999	1.307	147.692
Ethanol	lb/hr	11	6	5	5	5	-	5	5	0
Water	lb/hr	672	575	98	0	0	98	0	0	0
Ethylene	lb/hr	148.213	571	147.643	147.643	147.643	-	147.643	148	147.495



Oligomerization

[15]

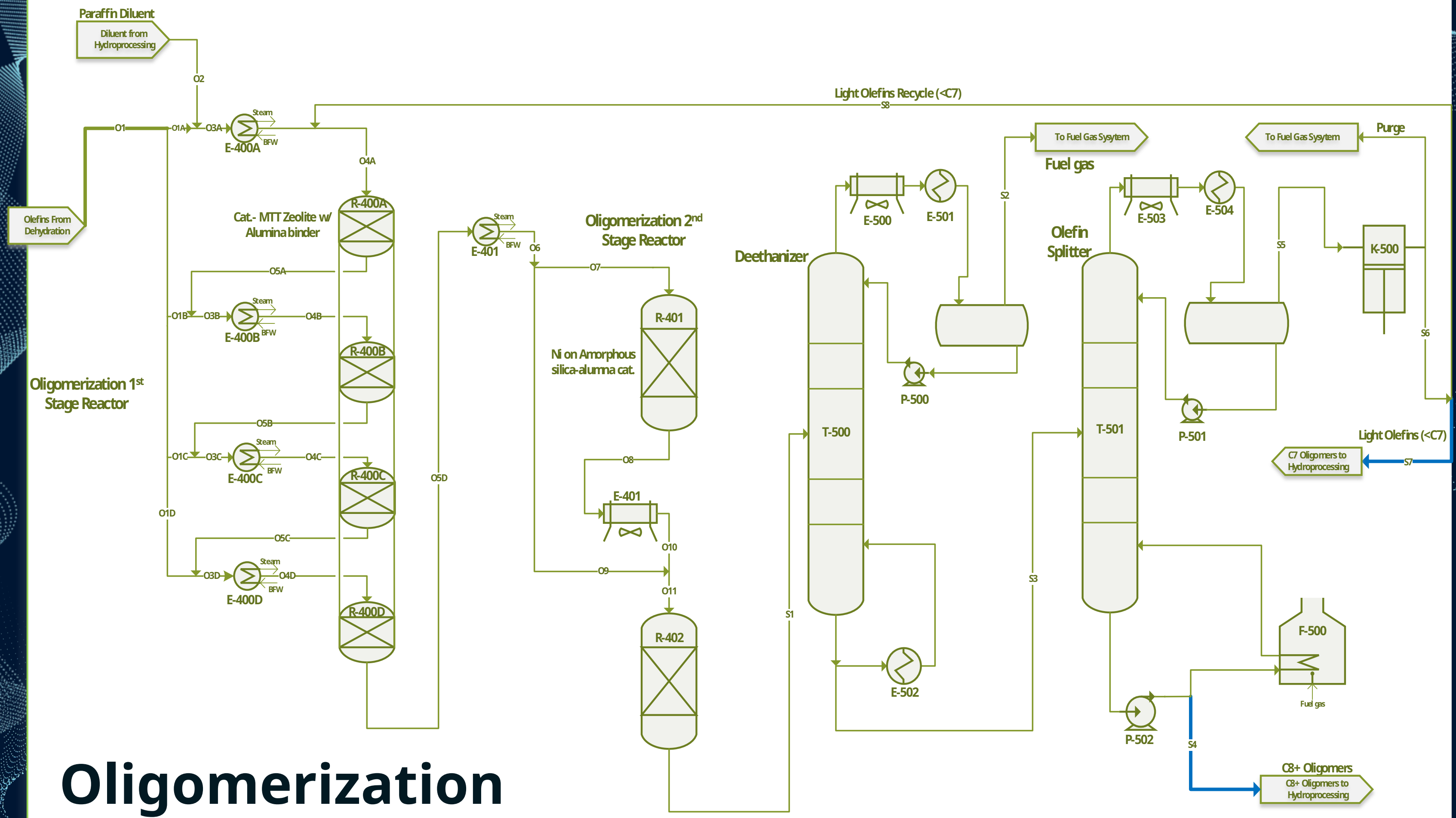


Oligomerization Reaction

Component	Units	O1	O1A	O2	O3A	O4A	O5A	O1B	O3B	O4B
Temperature	F	112	198	410	265	**248**	**320**	198	283	**248**
Pressure	psia	1,015	1,005	1,005	1,005	1,000	1,010	1,005	1,005	1,000
Mass Flows	lb/hr	147,692	46,141	36,874	64,451	64,451	64,451	46,141	110,592	110,592
Ethylene	lb/hr	147,495	36,874	0	36,874	36,874	549	36,874	37,423	37,423
Butene	lb/hr	0	0	0	4,097	4,097	39,678	0	39,678	39,678
Pentene	lb/hr	0	0	0	0	0	0	0	0	**0**
Hexene	lb/hr	21,806	0	21,806	21,806	30,567	0	30,567	30,567	**39,329**
Heptene	lb/hr	0	0	0	0	0	0	0	0	**0**
Octene	lb/hr	2,726	0	2,726	2,726	3,821	0	3,821	3,821	**4,916**

Component	Units	O6	O7	O8	O9	O10	O11	S1
Temperature	F	248	**248**	**320**	248	248	**248**	**320**
Pressure	psia	990	990	990	990	985	985	**985**
Mass Flows	lb/hr	202.875	101.438	101.438	101.438	101.438	202.875	**202.875**
C_2-C_7 Olefins	lb/hr	160.608	80.304	18.208	80.304	18.208	98.512	**36.415**
C_8-C_{17} Olefins	lb/hr	4.916	2.458	62.899	2.458	63.727	65.357	**125.798**
C_{18+} Olefins	lb/hr	-	-	1.655	-	-	1.655	**3.310**
Parraffins	lb/hr	36.874	18.437	18.437	18.437	18.437	36.874	**36.874**
Non-Condensables	lb/hr	198	98	98	98	98	198	**198**

[16]

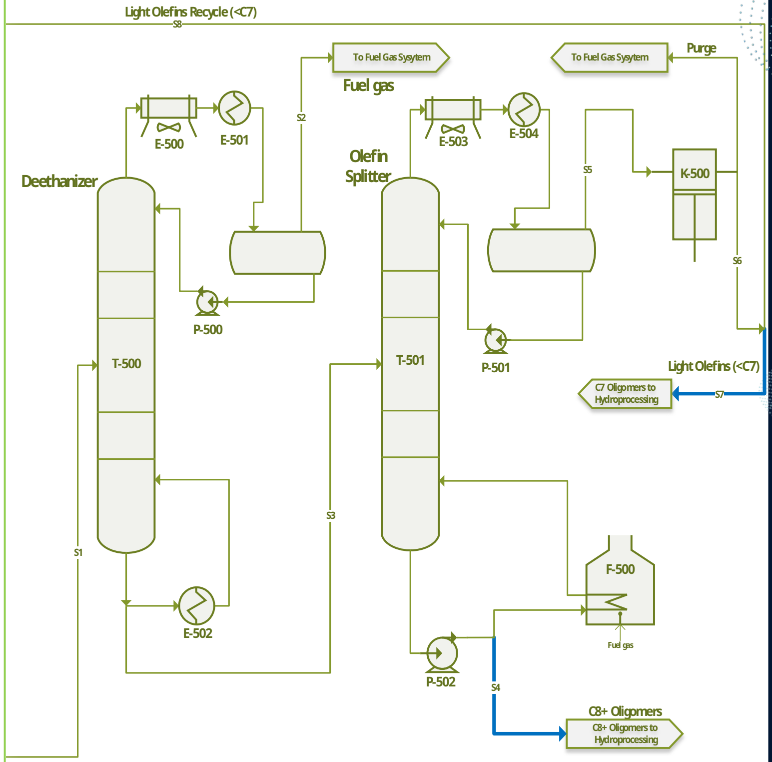


Oligomerization

[17]

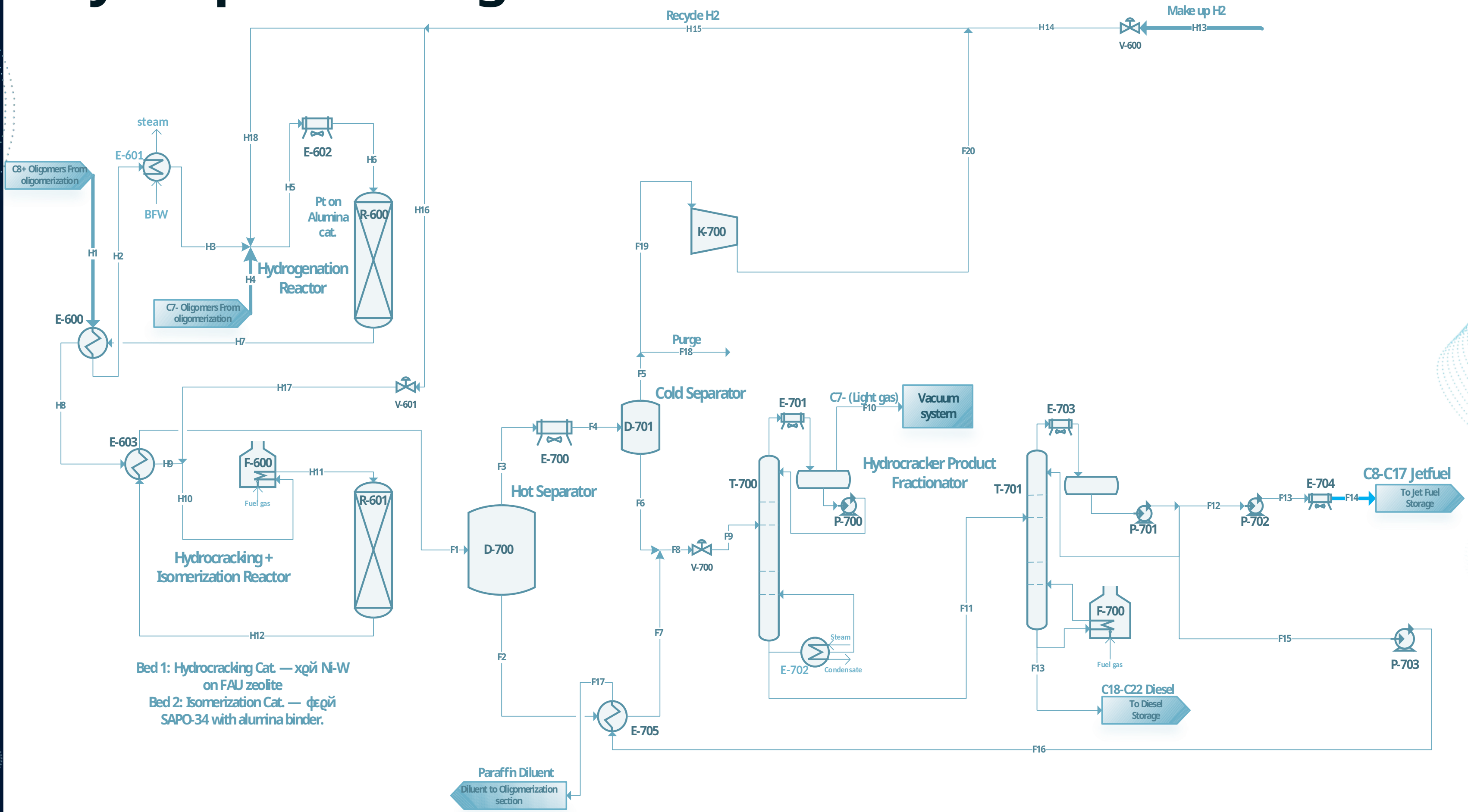
Component	Units	S1	S2	S3	S4
Temperature	F	320	19	600	394
Pressure	psia	985	415	417	36
Mass Flows	lb/hr	202,875	279	202,597	165,977
C_2-C_7 Olefins	lb/hr	36,415	82	36,333	**275**

Component	Units	S5	S6	S7	S8
Temperature	F	179	396	396	396
Paraffins	lb/hr	36,874	-	36,874	**36,874**
Pressure	psia	[illegible]	1,015	1,015	1,015
Non					
Mass Flows	lb/hr	36,619	36,619	**18,310**	18,310
C_2-C_7 Olefins	lb/hr	36,058	36,058	**18,029**	18,029
C_8-C_{17} Olefins	lb/hr	1	1	0	0
C_{18+} Olefins	lb/hr	0	0	0	0



Olefin Separation

Hydroprocessing



Bed 1: Hydrocracking Cat. — xọй Ni-W on FAU zeolite
Bed 2: Isomerization Cat. — фєọй SAPO-34 with alumina binder.

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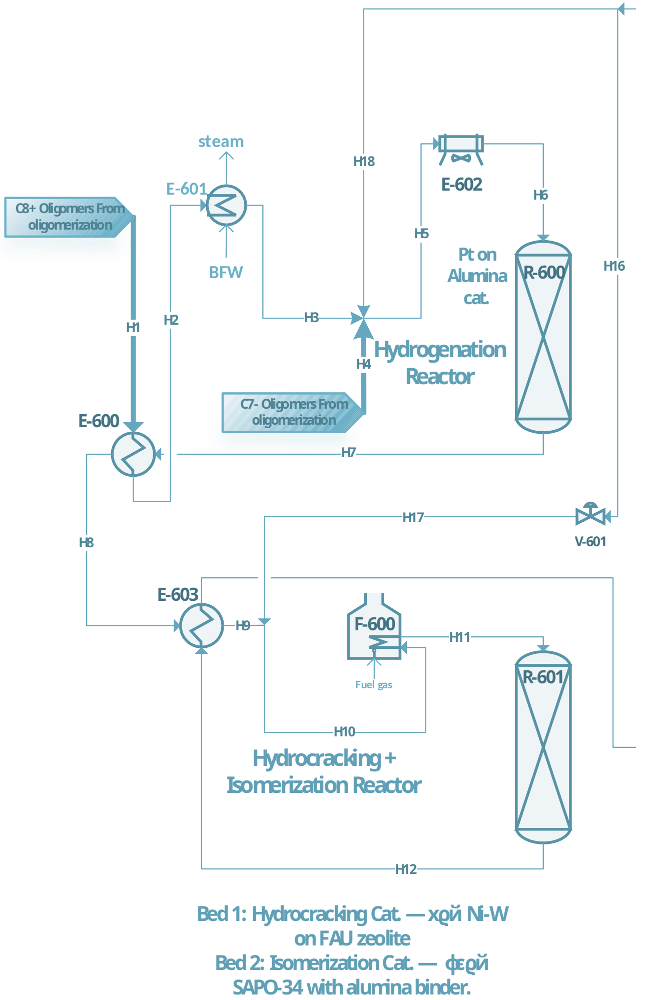
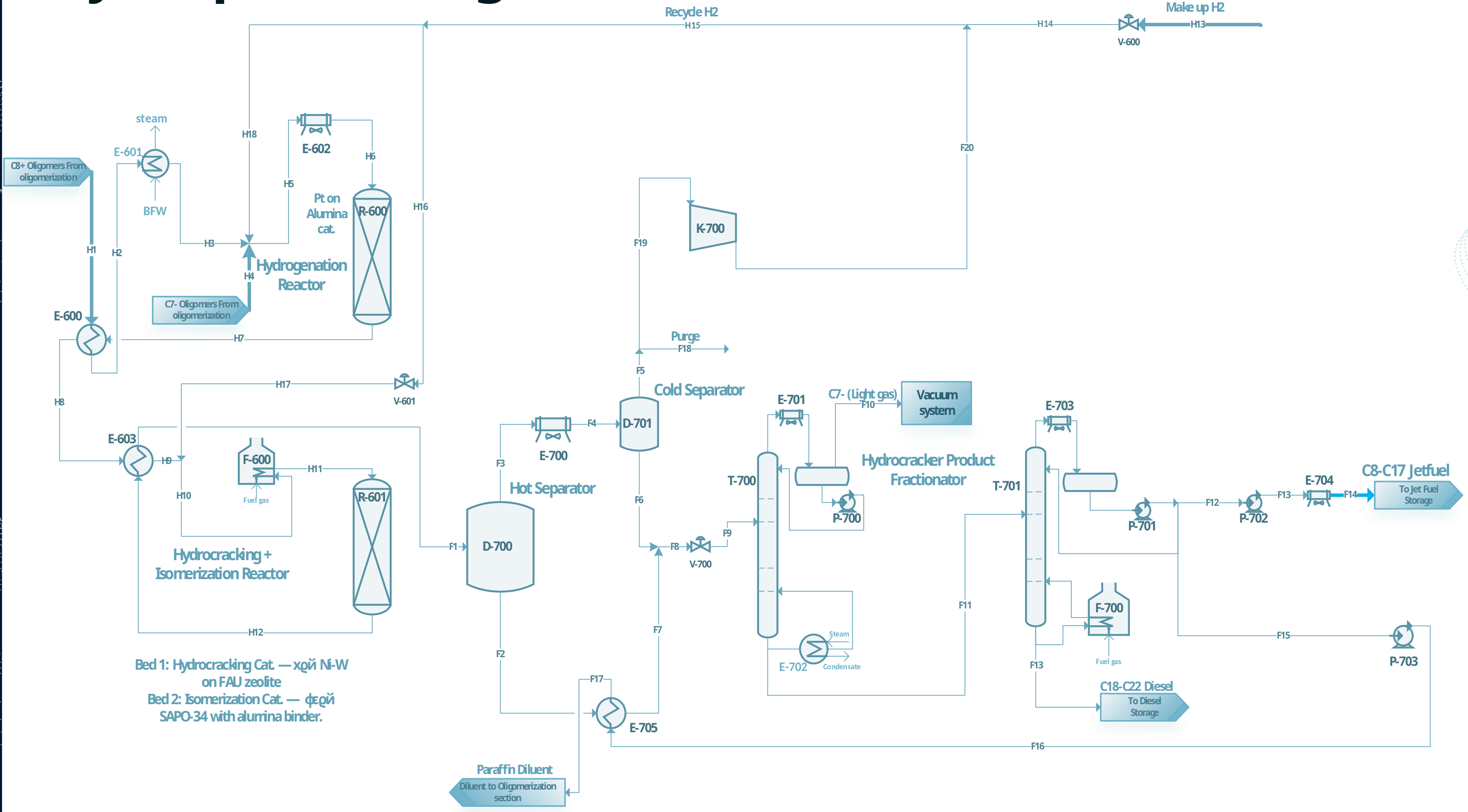
Hydroprocessing Reactions



Component	Units	H9	H10	H11	H12	H13
Temperature	F	716	685	752	748	257
Pressure	psia	909	904	899	834	2.176
Mass Flows	lb/hr	192.325	195.769	195.769	195.769	3.107
Hydrogen	lb/hr	3.484	5.927	5.927	5.605	3.107
C_2-C_7 Olefins	lb/hr	18	18	18	-	-
C_8-C_{17} Olefins	lb/hr	126	126	126	-	-
C_{18+} Olefins	lb/hr	3	3	3	-	-
C_2-C_7 Paraffins	lb/hr	21.432	22.342	22.342	19.016	-

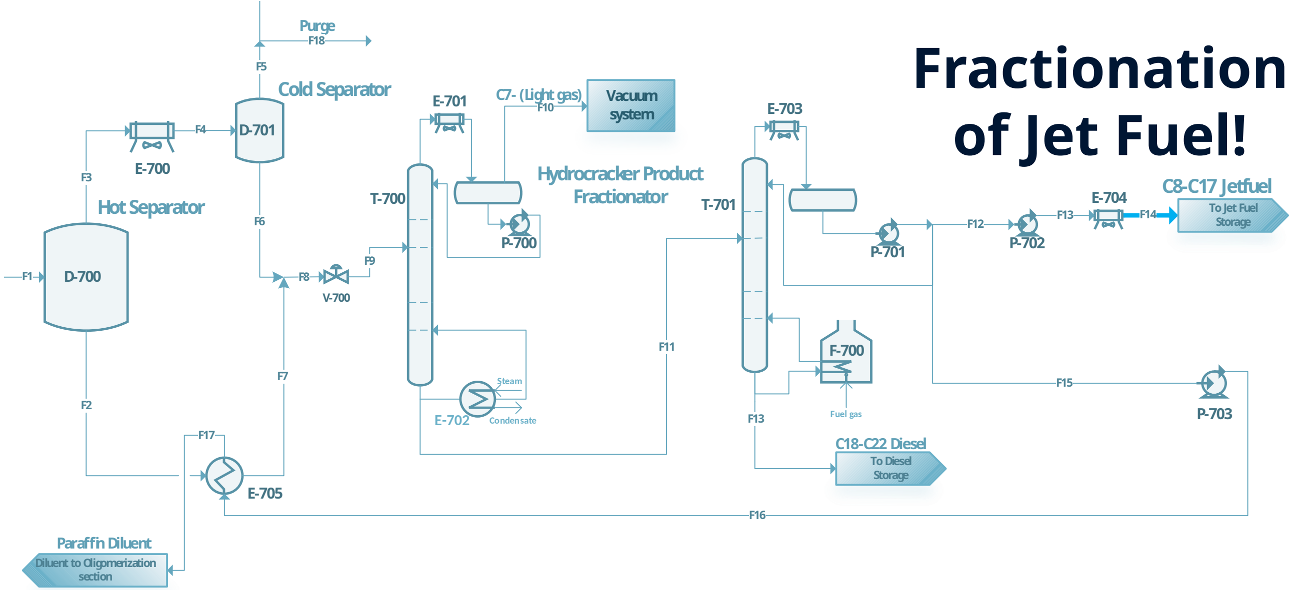
Component	Units	H14	H15	H16	H17	H18
Temperature	F	257	233	233	233	233
Pressure	psia	1.015	1.015	1.015	904	1.015
Mass Flows	lb/hr	3.107	11.482	3.445	3.445	8.038
Hydrogen	lb/hr	3.107	8.144	2.443	2.443	5.701
C_2-C_7 Olefins	lb/hr	-	-	-	-	-
C_8-C_{17} Olefins	lb/hr	-	-	-	-	-
C_{18+} Olefins	lb/hr	-	-	-	-	-

Hydroprocessing



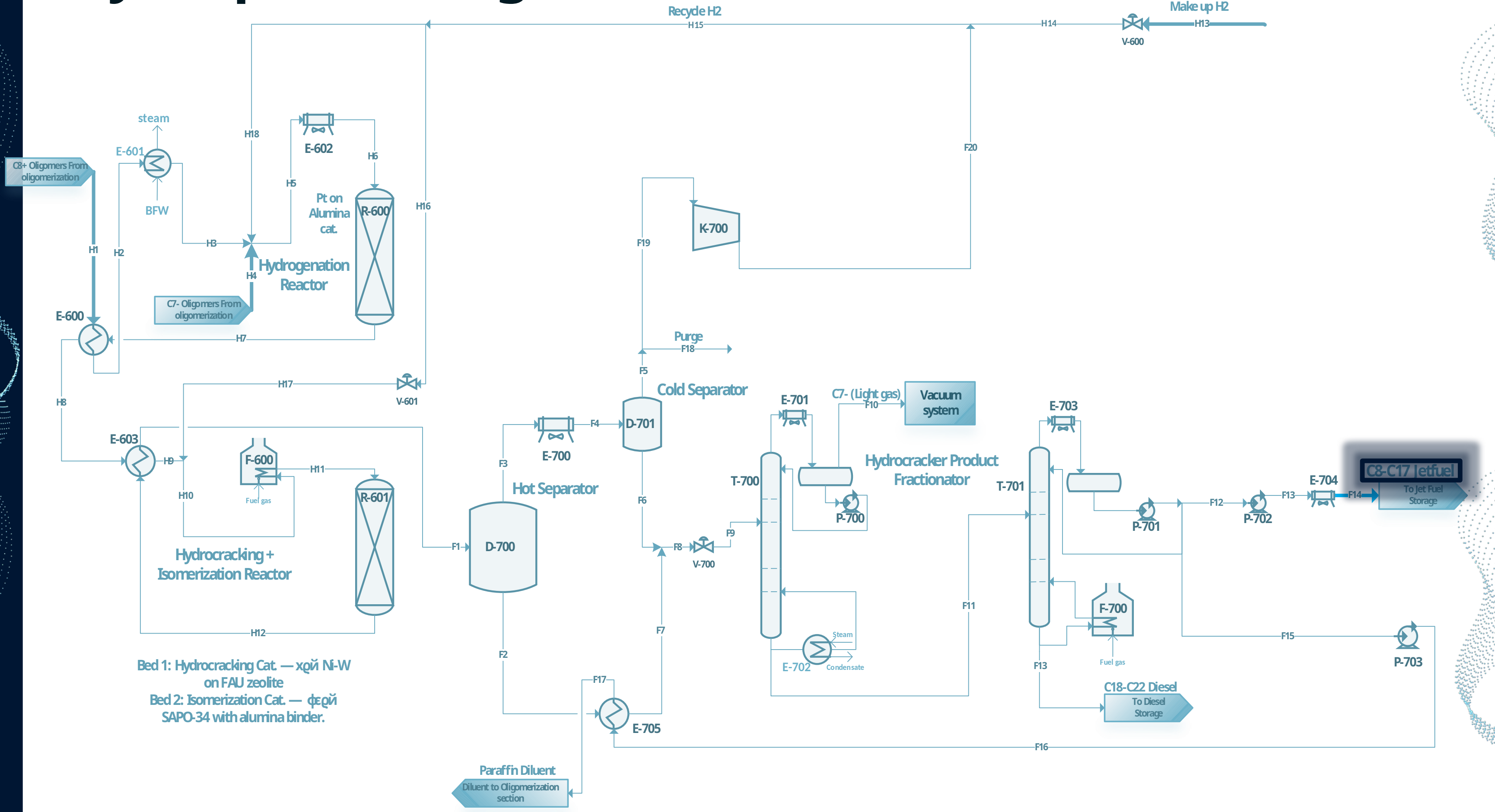
[21]

Fractionation of Jet Fuel!



Component	Units	F11	F12	F13	F14	F15	F16	F17	F18	F19	F20
Temperature	F	377	321	321	113	321	330	410	176	176	220
Pressure	psia	22	12	25	20	12	1,015	1,015	819	819	1,015
Mass Flows	lb/hr	173,037	116,854	116,854	116,854	36,874	36,874	36,874	931	8,375	8,375
Hydrogen	lb/hr	0	-	-	-	-	-	-	560	5,037	5,037
C_2-C_7 Olefins	lb/hr	-	-	-	-	-	-	-	-	-	-
C_8-C_{17} Olefins	lb/hr	-	-	-	-	-	-	-	-	-	-
[illegible] Olefins											

Hydroprocessing



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Overall Controls

Types of Equipment	Controls
Reactors	Temperature, Pressure, Flow, Material ratio controls
Heat Exchangers	Temperature, Flow
Pumps	Pressure, Flow
Compressors	Pressure, Flow
Driers	Regeneration Temp, Moisture Content
Drums	Pressure, Level
Columns	Temperature, Pressure, Flows, Level

Utility Summary

Name	Fluid	Rate	Rate Units	Cost/Year
Electricity	N/A	6,131	KW	**$2.9MM**
Cooling Water	Water	82	MMBTU/H	$0.24MM
Fuel Gas	Natural Gas	283	MMBTU/H	**$10.1MM**
LP Steam	Steam	4.6	MMBTU/H	$0.07MM
MP Steam	Steam	180	MMBTU/H	**$3.3MM**
HP Steam	Steam	11	MMBTU/H	$0.24MM
Refrigerant	Propane	10	MMBTU/H	$0.23MM

Total Utilities: $17.2MM/yr

03 | Process Economics

Cost and Value Analysis for the ETJ Process

Cost Analysis

CAPEX

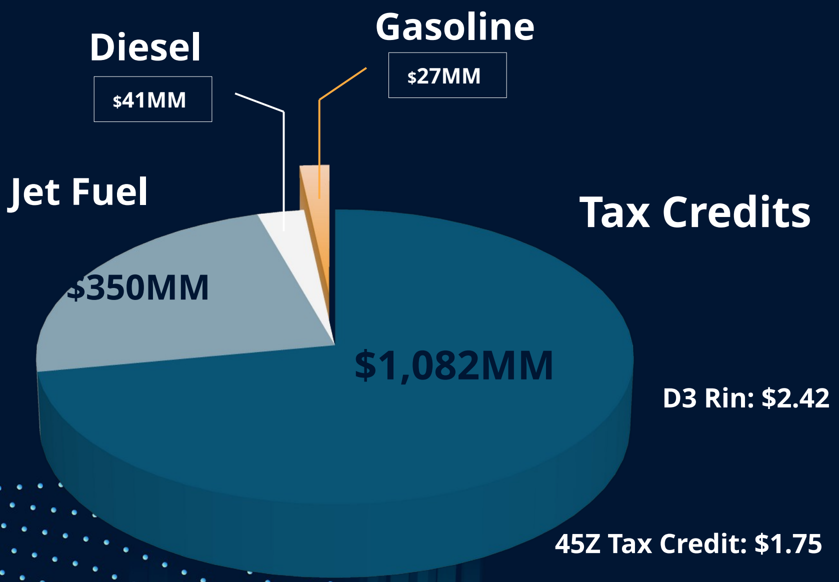
Capital Cost Evaluation	
ISBL	$95MM
OSBL (40% ISBL)	$38MM
ECC (10% ISBL)	$10MM
Contingency (10% ISBL)	$10MM
WC (15% ISBL)	$14MM
TCC	**$167MM**

OPEX

Cost of Production Evaluation	
FCOP	$10MM
Ethanol	**$518MM**
Hydrogen	$17MM
Utilities	$17MM
Catalysts	$5MM
Waste	$2MM
VCOP	**$560MM**
CCOP	**$570MM**

Income Breakdown

Revenue Streams



Revenue: $1,500MM/yr

Gross Proft: $930MM/yr

Net Proft: $687MM/yr

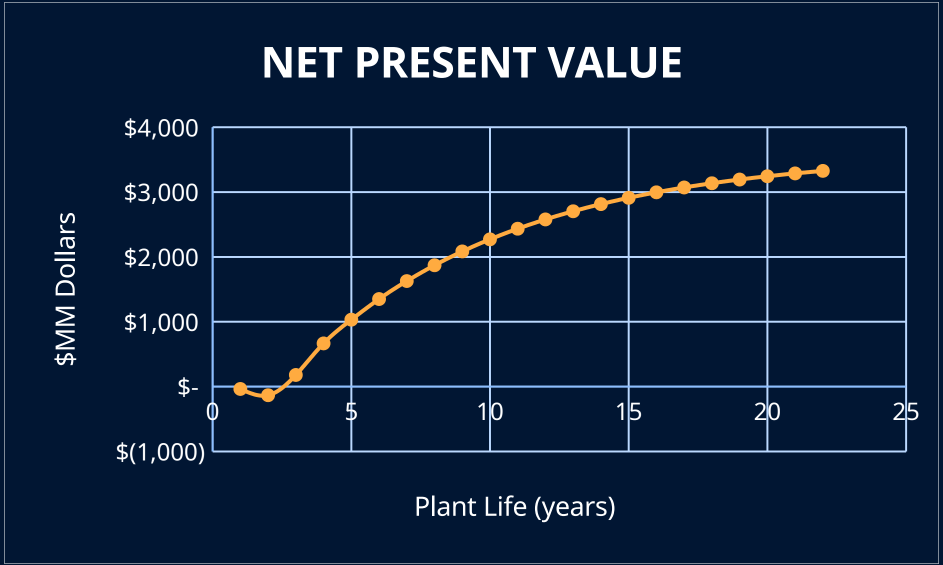
26% Net Tax Rate

Economic Evaluation Metrics

2.27yrs
SPBP

219%
DCFROR

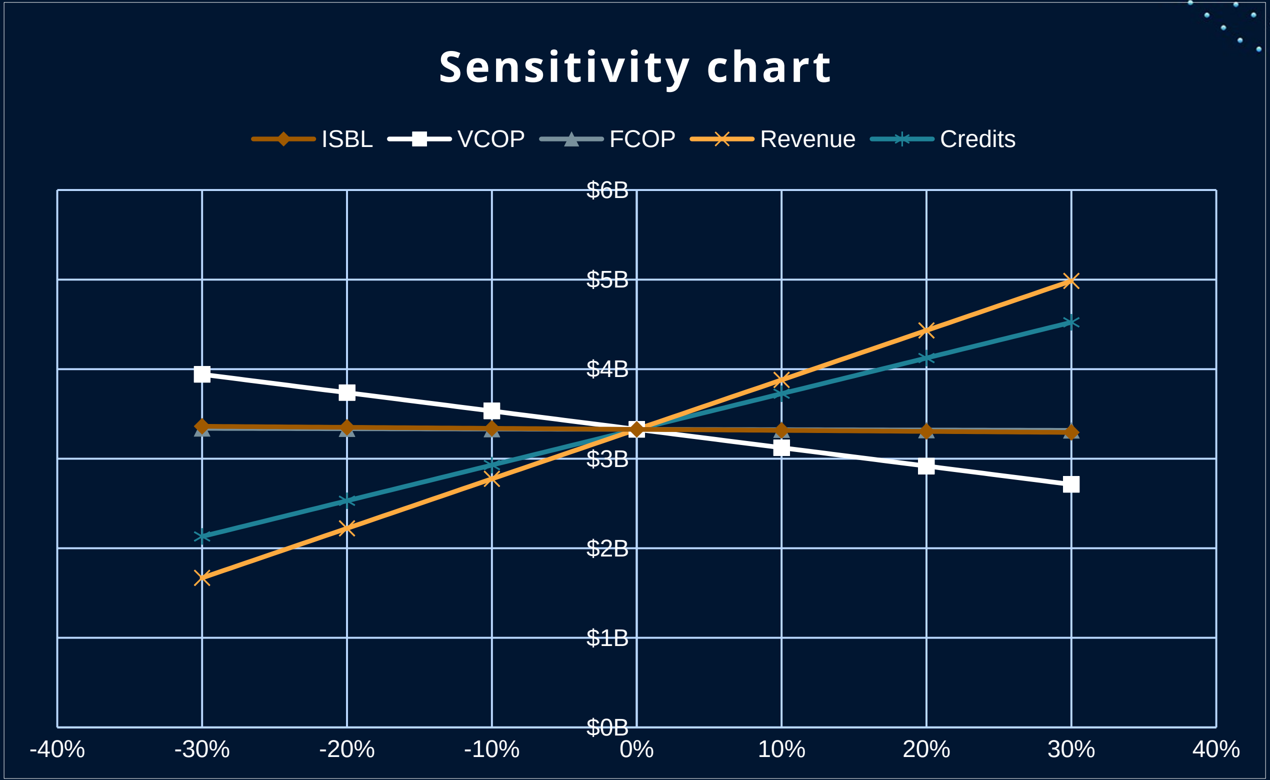
415%
ROI



$3,313MM
NPV at Year 20

[31]

Sensitivity Analysis



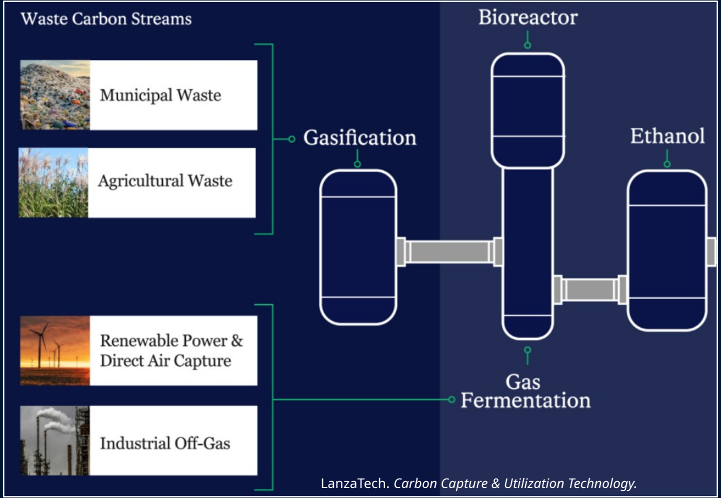
Biggest Factors:
Revenue & Credits

Smallest Factors:
FCOP & ISBL

04 Conclusion!

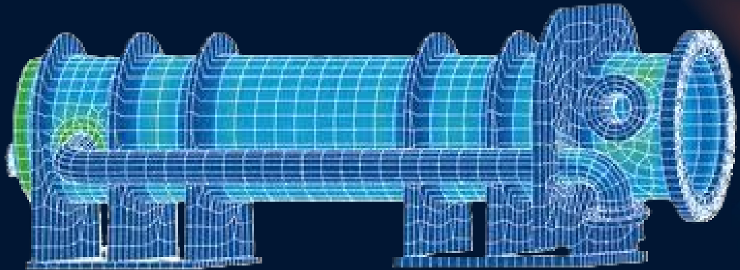
Recommendations

Produce Low-Carbon-Intensity Ethanol & Hydrogen



LanzaTech. *Carbon Capture & Utilization Technology.*

Heat Integration from Oligomerization to Dehydration


Quiri. *Skills – Complex Project Management.*

Increase Design Basis

Ultimate Cell. *How Do Electrolysers Work? A Complete Guide to Hydrogen Generation for Zero-Emission Vehicles*, 2025.

SPECIAL THANKS

LANCE BAIRD

Mentor

BETUL BILGIN

Advisor

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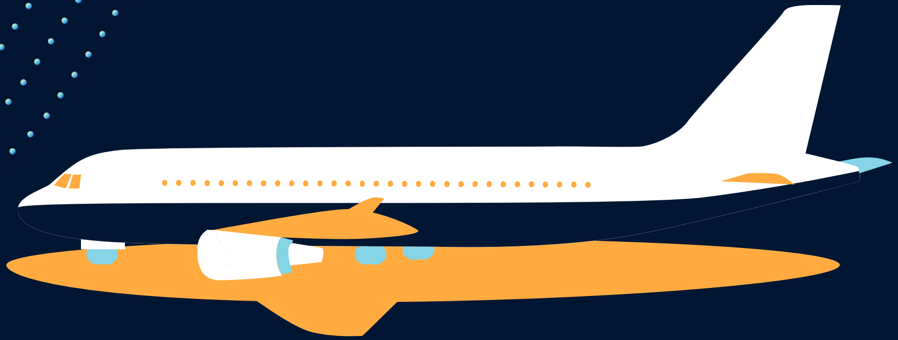


ANY QUESTIONS?

05 Additional Material

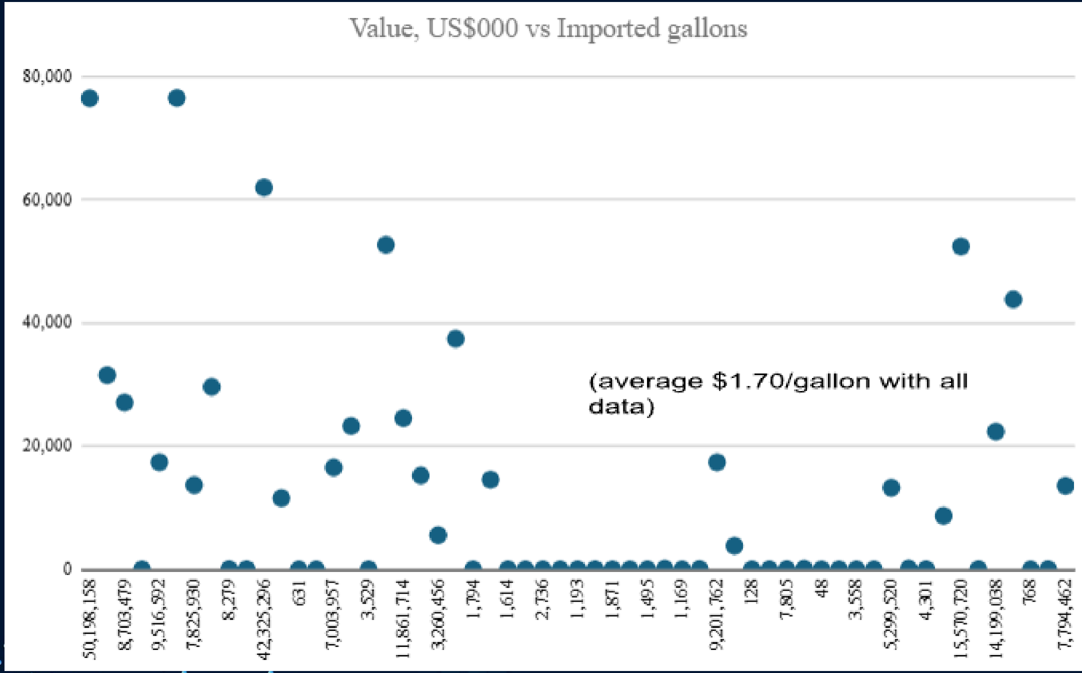
GHG Emissions For LanzaTech Ethanol

Table 3. GHG Emissions for LanzaTech Scenarios

	GHG emissions (g CO_{2eq}/MJ ethanol)			
scenario	BOF gas	corn stover	switchgrass	forest residue
net gas used[a]	−79.0	−80.2	−80.2	−80.2
feedstock procurement	0.0	11.2	14.9	6.6
utilities (heat and power)[b]	28.8	−6.3	−6.3	−8.1
other inputs[c]	1.8	1.3	1.3	1.3
anaerobic digestor emissions	6.8	7.5	7.5	7.5
waste treatment[d]	0.9	1.1	1.1	1.0
ethanol transport	0.7	0.7	0.7	0.7
ethanol combustion	71.4	71.4	71.4	71.4
net EtOH GHG emissions	31.4	8.0	11.7	1.5
% reduction[e]	67%	92%	88%	98%

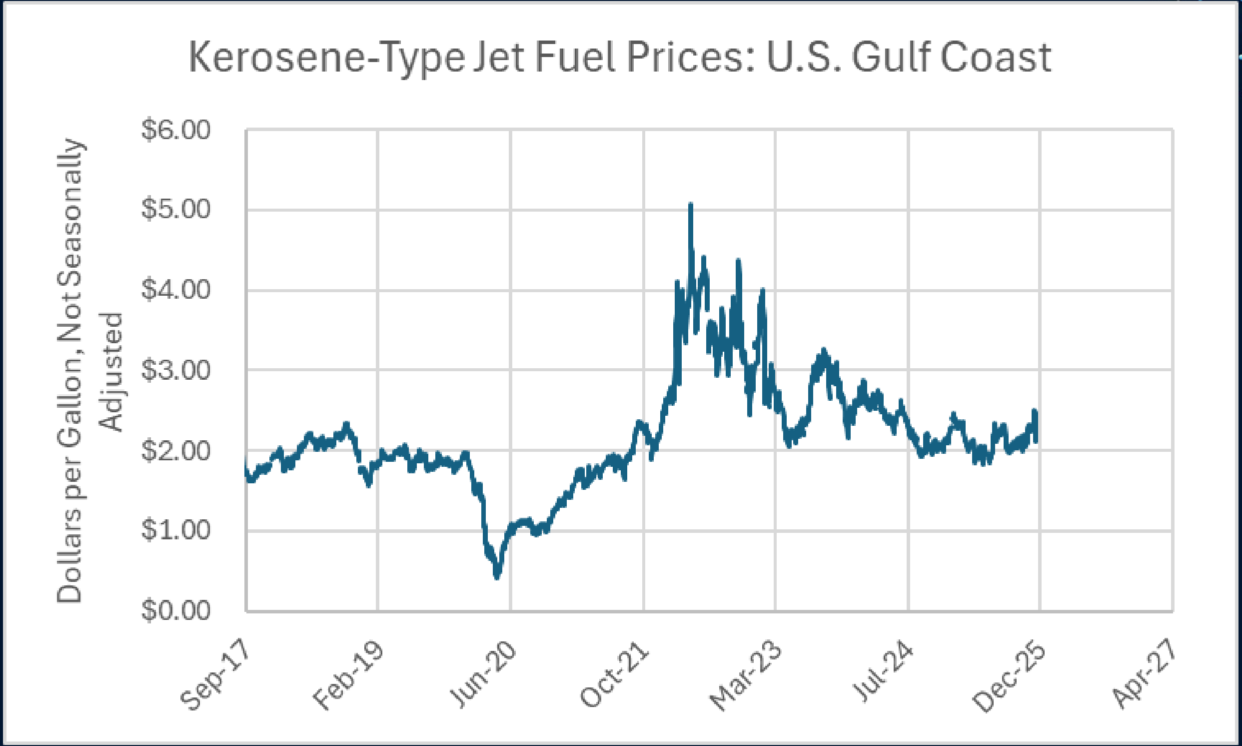
Retrieved from "Life Cycle Assessments of Ethanol Production via Gas Fermentation: Anticipated Greenhouse Gas Emissions for Cellulosic and Waste Gas Feedstocks."

Ethanol Price



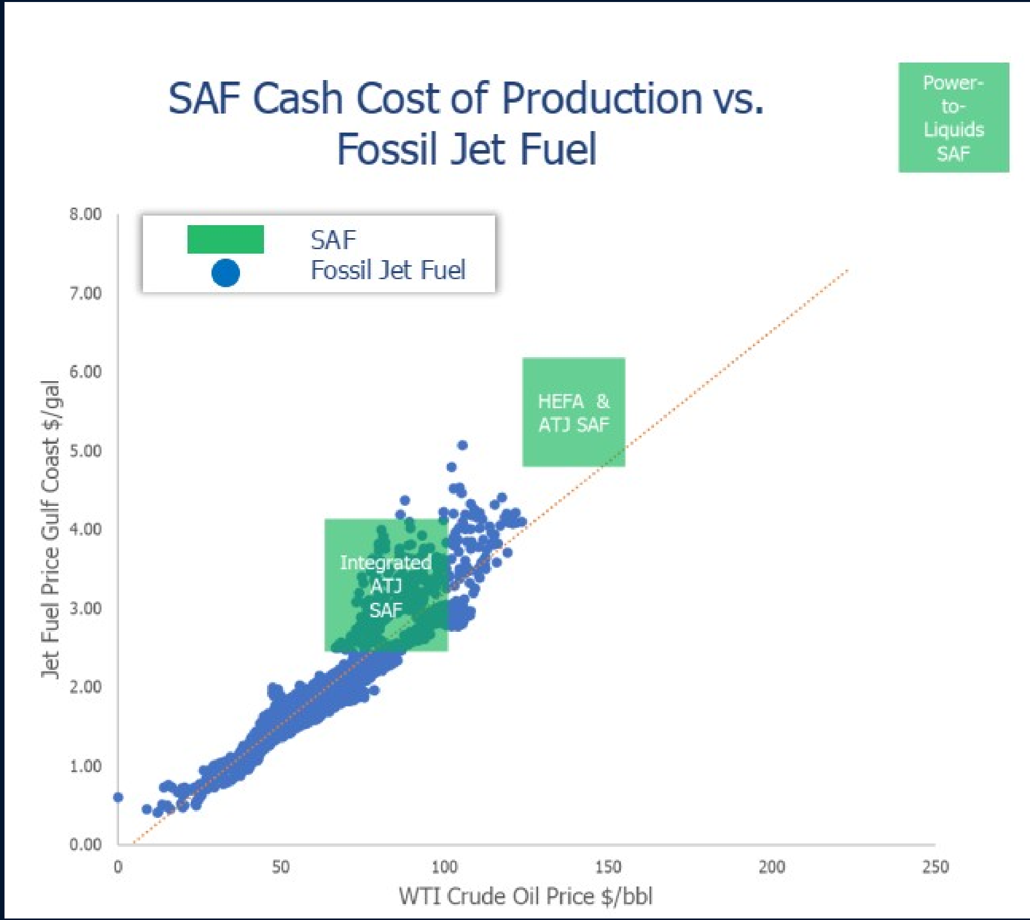
Data Source: U.S. Census Bureau Trade Data

Jet Fuel Price



U.S. Energy Information Administration, Kerosene-Type Jet Fuel Prices: U.S. Gulf Coast [DJFUELUSGULF], retrieved from FRED, Federal Reserve Bank of St. Louis; https://fred.stlouisfed.org/series/DJFUELUSGULF, March 15, 2026.

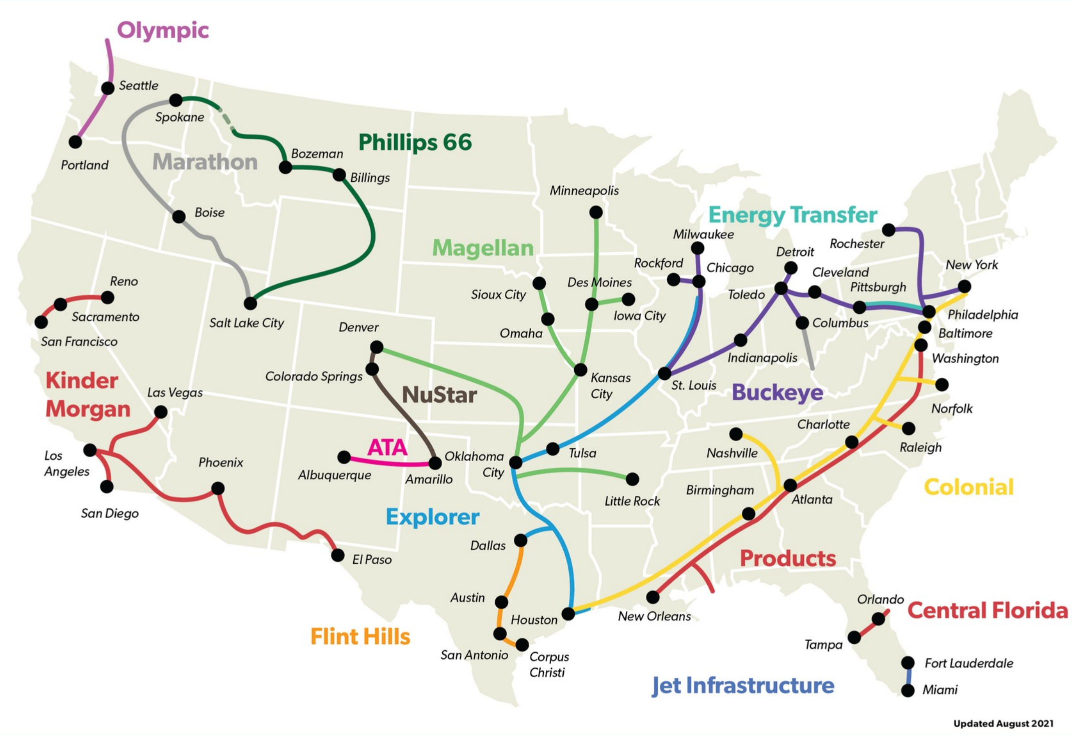
SAF vs. Fossil



Jet Fuel Pipelines

MAJOR U.S. PIPELINES CARRYING JET FUEL

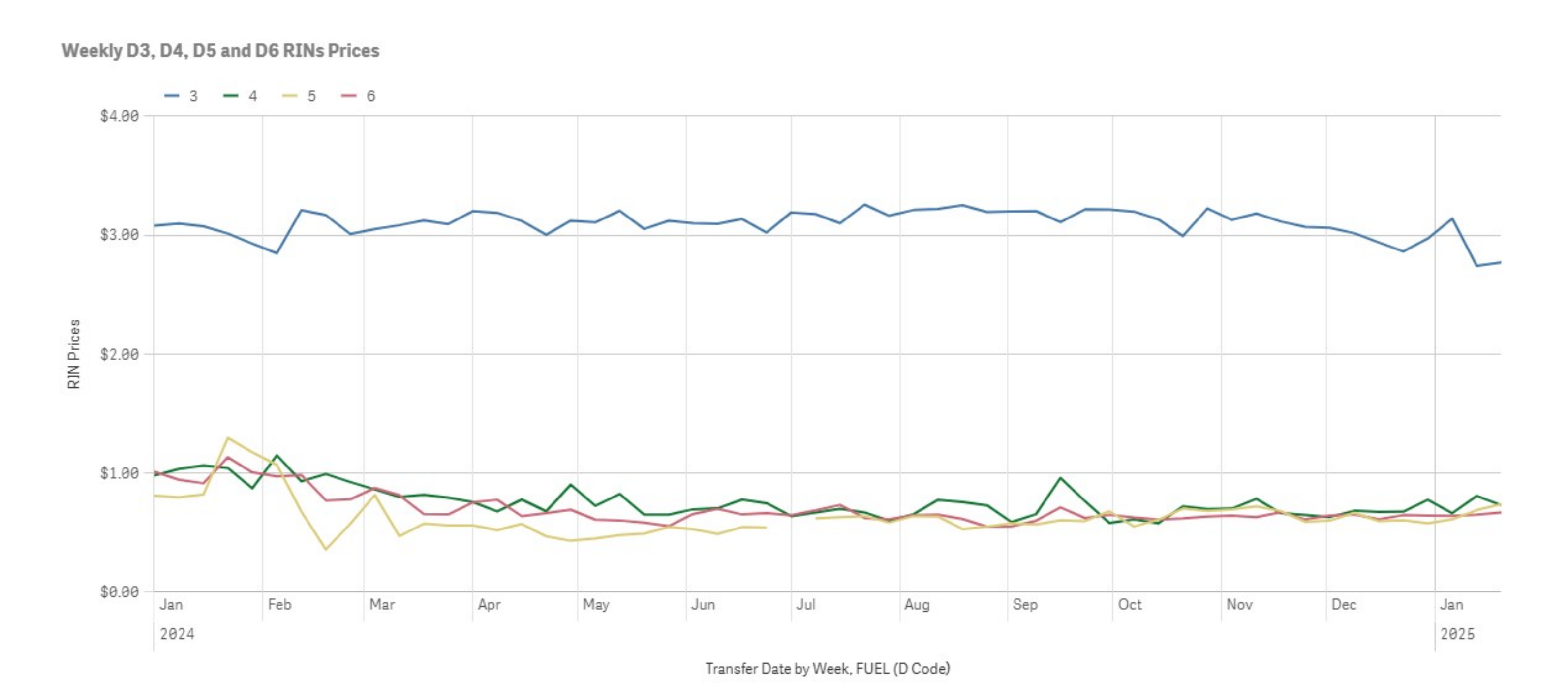
Most jet fuel is supplied to U.S. airports through a combination of multiproduct pipelines and dedicated local pipelines.



his map provides an overview of the major pipelines in the U.S. that move jet fuel. Smaller pipelines and cities supported indirectly by pipeline transport are equally important but are not shown on this map.

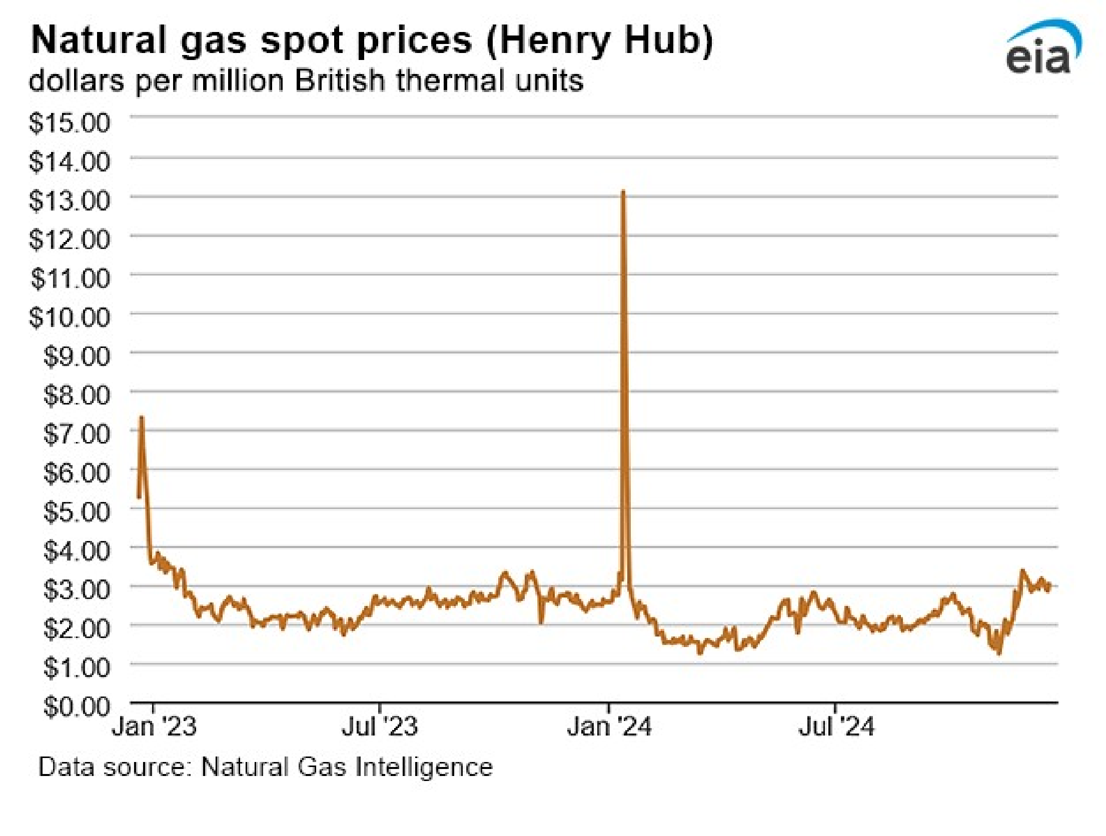
airlines.org

Rin Prices



Source: U.S. Environmental Protection Agency (EPA), “RIN Trades and Price Information,” retrieved March 15, 2026.

Natural Gas Price



Source: U.S. Energy Information Administration (EIA), Natural Gas Weekly Update.

Waste Pricing

MONTHLY WATER RATES		
USAGE TIER	INSIDE CITY RATE	OUTSIDE CITY RATE
Base Charge	$6.56	$6.56
>0-3 CCF*	$2.58 per CCF	$3.51 per CCF
>3-6 CCF	$5.34 per CCF	$6.48 per CCF
>6 CCF	$6.16 per CCF	$7.47 per CCF
Irrigation Meter	Same as tiered water rate charges otherwise applicable.	Same as tiered water rate charges otherwise applicable.
Wholesale**	$3.70	

MONTHLY SEWER RATES	
USAGE TIER	RATE
Base Charge	$6.56
>0-3 CCF*	$9.74 per CCF
>3-6 CCF	$13.64 per CCF
>6 CCF	$15.69 per CCF

Source: City of Atlanta Department of Watershed Management, "Water and Sewer Bill Rates," retrieved March 15, 2026.

Catalyst Costing

					2	beds						
	dehydratio n		**oligo- merization 1**		**Drier**		**olefin hydro- genation**		**Hydro- cracking**		**olig 2**	
bpd/SCFH	1,500		1,500		137,419		1,500		1,500		1,500	
m3/hr	142		19		-		33		137		16	
LHSV/GHSV	1		1		3,986		1		0.5		1	
catalyst	142	*m3*	19	*m3*	1,710	*lbs*	33	*m3*	273	*m3*	16	*m3*
density	680	*kg/m3*	680	*kg/m3*	49.6	*lb/ft3*	680	*kg/m3*	680	*kg/m3*	680	*kg/m3*
loading	96,693	*kgs*	12,785	*kgs*	34	*ft3*	22,406	*kgs*	185,823	*kgs*	10,674	*kgs*
	213,208	*lbs*	28,190	*lbs*	776	*kgs*	49,405	*lbs*	409,740	*lbs*	23,536	*lbs*
lifespan	3	*years*	3	*years*	3	*years*	3	*years*	3	*years*	3	*years*
consumption	71,069	*lbs/year*	9,397	*lbs/year*	259	*lbs/year*	16,468	*lbs/year*	136,580	*lbs/year*	7,845	*lbs/year*
Price	$10	*/lb*	$15	*/lb*	$15	*/lb*	$20	*/lb*	$25	*/lb*	$15	*/lb*
rate	**$710,694**	*/year*	**$140,952**	*/year*	**$3,879**	*/year*	**$329,367**	*/year*	**$3,414,502**	*/year*	**$117,682**	*/year*
cost/load	$2,132,081		$422,857		$11,636		$988,102		$10,243,505		$353,047	

HEFA Process

Feedstock: Oil-based products (FOGS)

Major Licensors: UOP, Neste, Axens, & Haldor-Topsoe,

Advantages: Utilizes oil & grease, simpler* process pathway, most industrialized

Disadvantages: Feedstock is food, limited quantity, and expensive

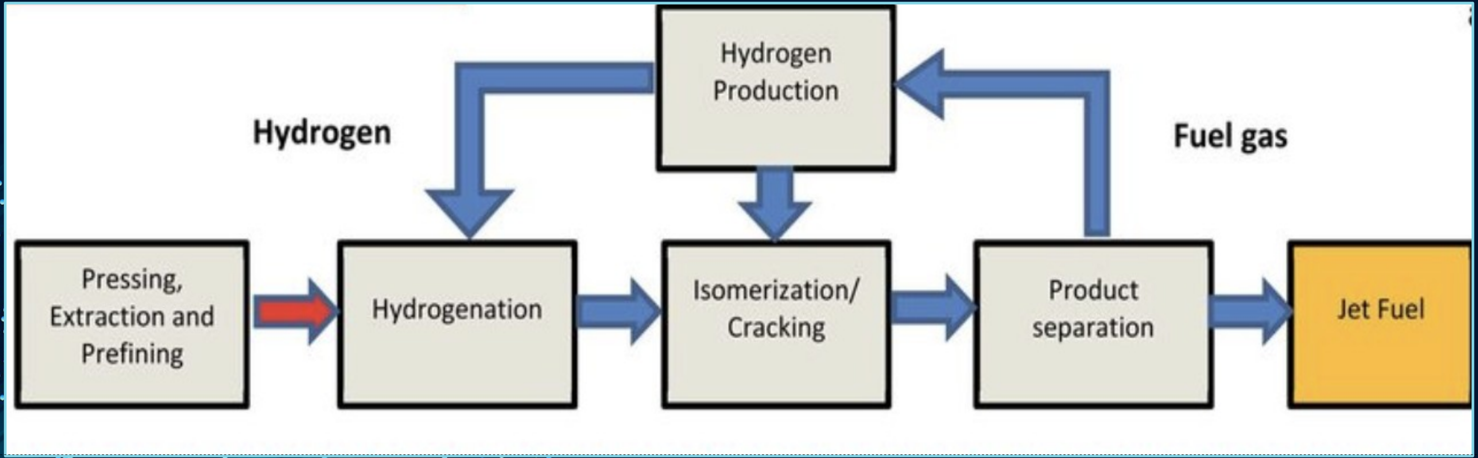


Figure 1. *Overview of the HEFA (hydro processed esters and fatty acids) pathway*

Source: *Adapted from the HEFA schematic in "Overview of HEFA, Fischer–Tropsch, and sugar-to-jet pathways," as reproduced on ResearchGate (Figure 2).*

Gasification/Fisher Tropsch

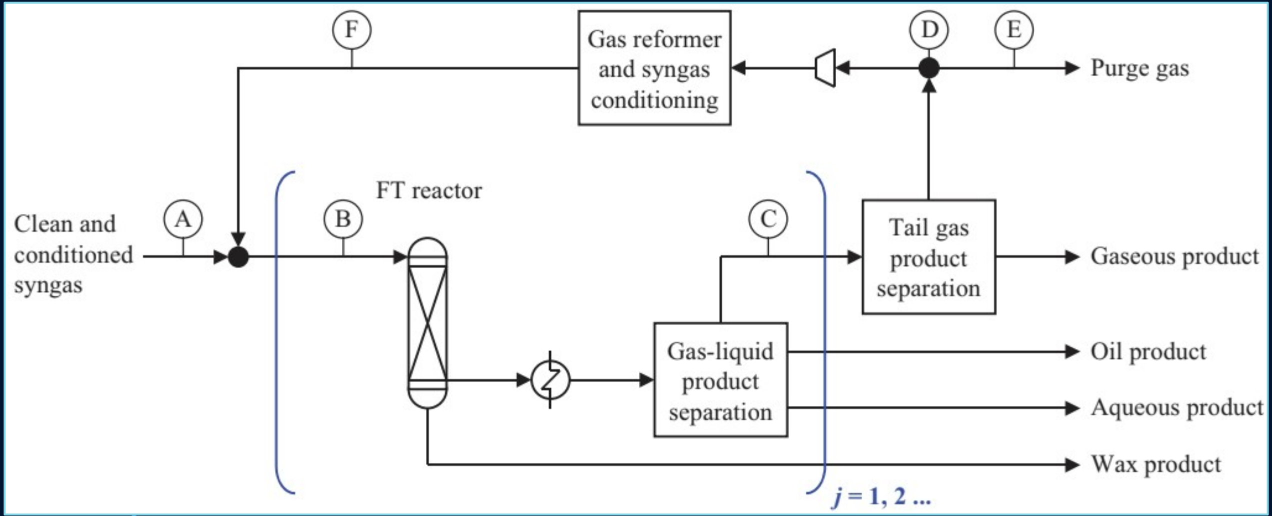


Figure 2. *Simplified Fischer–Tropsch gas loop for low-temperature FT (LTFT) showing clean syngas feed (point A), FT reactor*
Source: Adapted from de Klerk, A., "Fischer–Tropsch process," in Kirk-Othmer Encyclopedia of Chemical Technology, John Wiley & Sons, [2013].

Feedstock: Biomass

Major Licensors: Fulcrum Bioenergy, VLS, & Rentech.

Advantages: High diversity of feedstock, wide range of products,

Disadvantages: Complicated process, high CAPEX and OPEX, co-products produced

Pyrolysis/Hydrotreating

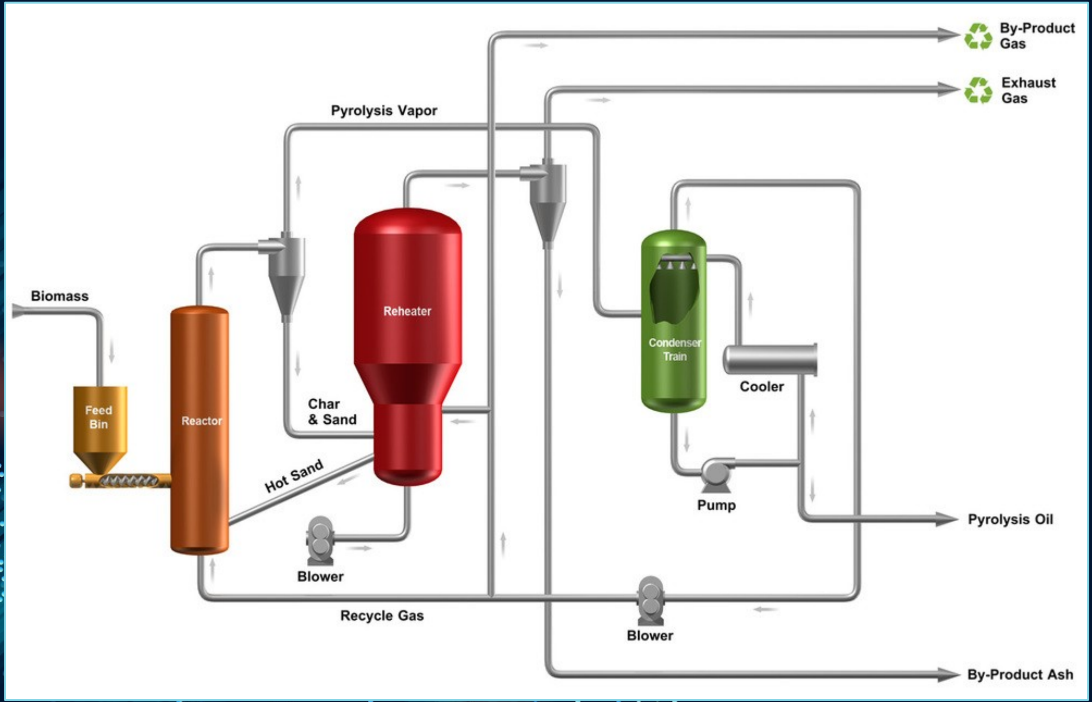


Figure 3: *Fast Pyrolysis Process*
Source: *Ensyn Corporation, "Technology Overview: RTP® Process," 2015.*

Feedstock: Waste feedstock biomass

Major licensors: Envergent, Honeywell

Advantages: Handles diverse feedstocks & low cost of production

Disadvantages: Unstable pyrolysis oil, reactor plugging and high catalyst cost

WHY ETHANOL TO JET?

Flexible Feedstock	Green ethanol can be derived from corn, sugarcane, waste & **more!**
High SAF Selectivity	Up to **90% of hydrocarbons** produced are in jet fuel range!
Clean Process	No char, tar, or other hard to handle co-products.
Proven Procedure	Recently commercialized and **ASTM certified** by LanzaJet!